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Absorbing or externalizing uncertainty? Nuclear energy governance architectures and delivery regimes across global reactor builds

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ABSTRACT

Long construction lead times are a principal driver of nuclear cost escalation and delivery underperformance. Recent research increasingly identifies governance dynamics—including regulatory evolution, institutional instability, and procedural reopening—as important influences on infrastructure delivery. Yet despite shared reactor technologies and similar external shocks, nuclear construction performance varies markedly across countries and historical periods. What remains insufficiently explained is why some governance systems sustain comparatively stable construction trajectories while others experience persistent delay and disruption.

This study examines how governance architectures shape the capacity of nuclear delivery systems to adapt under conditions of uncertainty. Drawing on Transaction Cost Theory, the study employs an explanatory sequential mixed-methods design combining regression analysis of 541 completed commercial nuclear reactors worldwide (1954–2020) with a structured comparative analysis of major nuclear states. The quantitative analysis identifies associations between macro-institutional conditions and construction lead times, while the qualitative analysis explains how these broader institutional environments translate into distinct governance arrangements.

The findings suggest that differences in nuclear construction performance reflect recurring governance functions shaping system-level adaptive capacity over long implementation horizons. Shorter construction durations are associated with governance arrangements that sustain coordinated adaptation, cumulative learning, stable financing, and durable political authorization, whereas more fragmented and procedurally reopenable systems are associated with greater exposure to renegotiation and disruption. By linking macro-institutional conditions to recurring governance mechanisms, the study advances a governance-centered explanation of variation in nuclear infrastructure delivery and provides a comparative framework for understanding how institutional design shapes complex low-carbon megaprojects.

1. Introduction

Few domains illustrate the challenge of delivering capital-intensive low-carbon infrastructure as clearly as nuclear power. Intended to provide abundant, reliable electricity and strengthen long-term energy security, nuclear new-build projects have nevertheless become emblematic of persistent delays and cost escalation across many advanced economies. Average construction lead times have increased substantially since the 1970s, and emblematic projects—such as Olkiluoto-3, Flamanville-3, and Vogtle-3/4—have experienced decade-scale delays [2,39,43,53]. Likewise, a substantial empirical literature documents persistent cost escalation in nuclear power. Capital costs have risen over time, and overruns and schedule delays are widespread at the level of individual plants [45,46].

This has directed attention toward the determinants of project duration itself. Long construction lead times are widely recognized as a central driver of nuclear performance and cost escalation [8]. Existing techno-economic explanations have attributed prolonged construction to a range of drivers, from disrupted learning-by-doing associated with rising technical complexity [32], design proliferation, and lack of standardization [8,18], to the growing contribution of "soft costs," financing conditions, and safety-related requirements [14]. From a megaproject management perspective, nuclear construction exhibits institutional-sensitive characteristics common to transition-critical infrastructure: extreme asset specificity, multi-decade time horizons, and dense interdependence among utilities, vendors, regulators, financiers, and political authorities [16,34]. Yet the persistence of these patterns remains only partially understood.

Recent evidence places these patterns in a broader cross-technology perspective and makes governance central to explanation. Sovacool and

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Ryu show that, across major electricity-generation technologies, nuclear power exhibits the highest cost escalation, the longest construction durations, and the greatest variance in outcomes [47]. The authors show how ongoing "regulatory ratcheting", whereby safety standards and oversight evolve over time, repeatedly reshapes project through design modifications, approval reopening, and regulatory revision that amplify delay and cost escalation. While project-level execution matters across infrastructures, their analysis identifies nuclear as a distinctive case in which governance, regulatory, and procedural dynamics remain structurally central determinants of delivery performance [47,50].

This evidence indicates that nuclear underperformance cannot be understood as a purely technical or managerial problem but as a heavily governance-mediated outcome shaped by institutional exposure to uncertainty. Nuclear projects are highly sensitive to how authority, regulation, and coordination are organized within governance architectures over time. Nuclear construction is therefore not simply difficult to build technically, but institutionally fragile under conditions of uncertainty. However, these structural characteristics alone do not explain why similar disruptions translate into persistent delay and volatility in some countries but not in others.

Even in a sector highly sensitive to governance failure and regulatory ratcheting, performance does not converge uniformly toward failure. Across historical periods, some countries have sustained comparatively faster and more stable construction trajectories, indicating that adverse outcomes are not inevitable under similar technological conditions. Critically, cross-national differences persist despite technological diffusion and accumulated industry experience, which points less to a universal nuclear megaproject syndrome [34] than to systematic variation in governance arrangements across long-horizon infrastructure delivery under uncertainty.

For example, Budzier and Maylor document the refurbishment of the Darlington nuclear station in Canada as a successful case of deliberate governance orchestration and system-level program management [10]. Their analysis highlights how an "owner-as-integrator" model, integrated risk governance, and sustained coordination across regulators and supply chains can stabilize delivery in a sector often described as structurally prone to delay and overrun. This evidence suggests that nuclear performance is not uniformly pathological but contingent on how governance is organized and enacted over time.

What remains insufficiently explained is why such stabilizing arrangements emerge and persist in some countries but not in others. While we know governance is central, we lack a systematic explanation of how different governance architectures generate distinct and persistent construction performance regimes across national settings. If nuclear construction is universally exposed to technological complexity, regulatory ratcheting, and long investment horizons, why have some countries consistently delivered projects relatively quickly and predictably while others have experienced chronic delay, renegotiation, and cost escalation? More specifically, under what conditions are disturbances—regulatory change, financing shocks, or political contestation—absorbed through coordinated adaptation, and when are they translated into delay and instability?

This paper addresses this puzzle by asking how governance architectures shape the capacity of nuclear delivery systems to absorb uncertainty rather than externalize it into disruption. To answer this question, the study combines a transaction-cost perspective and institutional analysis [55,58] with a mixed-methods empirical design. Building on the governance-centered diagnosis advanced by Sovacool and Ryu [47], the quantitative strand examines cross-national variation in construction lead times and outcome variance across 540 reactors to identify statistical patterns consistent with broader macro-institutional environments shaping nuclear governance and adaptive capacity. Controlling for reactor vintage through decade fixed effects, region, major accident periods, and construction suspension dynamics, the quantitative analysis identifies robust associations between nuclear construction lead time and two macro-institutional conditions.

The qualitative stage then examines the mechanisms underlying these statistical relationships by tracing how broader macro-institutional environments translate into distinct governance regimes associated with different construction performance trajectories. A comparative analysis organized through performance clustering across nine major nuclear-building states examines how contrasting governance architectures shape coordination, learning processes, risk allocation, and political stabilization over time. Rather than treating governance as a singular institutional attribute, the analysis identifies interdependent governance complexes through which adaptive capacity is generated and sustained across different delivery regimes. Taken together, these findings contribute to a more nuanced understanding of how governance architectures shape infrastructure delivery under conditions of uncertainty in large-scale energy systems, with nuclear power serving as an analytically revealing infrastructure case.

2. Governance architectures, transaction costs, and nuclear construction under uncertainty

2.1. Infrastructure delivery and governance under uncertainty

A substantial body of research on the megaproject syndrome argues that the chronic cost overruns, schedule delays, and governance failures associated with large-scale infrastructure reflect not only technical inefficiency, but also deeper institutional and organizational dynamics [16,17,34]. Megaprojects—such as airports, dams, high-speed rail networks, and nuclear power plants—typically involve large capital requirements, long implementation periods, high political visibility, and extensive coordination across public and private actors. Existing research attributes underperformance to factors including optimism bias, scope change, fragmented coordination, institutional conflict, and governance failure [17,34]. Across these perspectives, a recurring concern is that infrastructure delivery becomes increasingly difficult when authority structures, coordination mechanisms, and risk-management arrangements become fragmented.

Within this broader class, nuclear power represents an especially coordination-intensive form of infrastructure [18,49]. Reactor projects are indivisible, site-specific, capital-intensive, and dependent on long-term regulatory, financial, and industrial coordination. Their implementation horizons—often extending beyond a decade—mean that disruptions in licensing, financing, regulation, or supply-chain coordination can cascade across the project system. Nuclear construction therefore combines technological complexity with sustained exposure to institutional and regulatory uncertainty.

Recent cross-technology evidence increasingly identifies governance as an important dimension of infrastructure delivery performance. Nuclear projects exhibit the highest cost escalation, the longest construction durations, and the greatest variance in outcomes across electricity-generation technologies, while evolving safety standards and oversight repeatedly reshape project specifications during construction through design modifications, approval resets, and procedural reopening processes [47]. Such processes of "regulatory ratcheting" suggest that nuclear construction is not only technically demanding, but also institutionally exposed to recurrent disturbance over long implementation horizons. More broadly, state capacity, market organization, and international coordination arrangements shape the feasibility and trajectory of nuclear deployment beyond technological characteristics alone [9].

Yet outcomes do not converge uniformly toward failure. Some countries sustain comparatively faster and less variable delivery trajectories, suggesting that performance may depend not only on technological conditions but also on how governance architectures organize coordination and adaptation over time [34]. Recent evidence similarly suggests that integrated program management, coordinated authority, and sustained interface management may stabilize delivery even in highly complex settings [10]. What remains insufficiently explained is

why such stabilizing arrangements emerge in some contexts but not others, and under what conditions disturbances are absorbed through coordinated adaptation rather than becoming associated with renegotiation, delay, and instability. In other words, the unresolved issue concerns how governance architectures organize adaptation under conditions of uncertainty.

2.2. Governance architectures and coordinated adaptation: A transaction cost perspective

Nuclear construction has also been examined through megaproject management, political economy, infrastructure studies, and socio-technical transition perspectives, among others. These approaches emphasize the importance of state capacity, institutional legitimacy, industrial coordination, and political contestation in shaping infrastructure delivery outcomes. Rather than replacing these perspectives, Transaction Cost Theory (TCT) is used here as a complementary analytical lens for interpreting how governance architectures organize coordinated adaptation under conditions of uncertainty and interdependence [11,57].

From this perspective, TCT provides a useful framework for interpreting variation in nuclear construction performance by focusing on how governance architectures organize coordinated adaptation and safeguard relationship-specific investments under uncertainty [11,55,57,59]. Under bounded rationality, complex transactions cannot be fully specified ex ante, and project performance depends heavily on the capacity to coordinate adaptation as conditions evolve. As uncertainty increases, contractual gaps widen and require ongoing adjustment; where such architectures lack sufficient governance capacity to coordinate adjustment internally, disturbances may increasingly translate into renegotiation, dispute, and procedural reopening [59].

Three transaction attributes are particularly relevant for nuclear construction: asset specificity, uncertainty, and interdependence. First, site-specific investments in design, licensing, and specialized supply chains create strong bilateral dependencies once investments become sunk, increasing exposure to hold-up and coordination problems [56,58]. Second, long construction horizons expose projects to evolving safety standards, financing conditions, regulatory revision, and political contestation, generating adaptive requirements that are difficult to fully anticipate through contractual arrangements alone [55]. Third, reactor delivery requires tightly coordinated interaction across technical, regulatory, financial, and political actors operating with different incentives and information asymmetries [59].

These conditions suggest that nuclear construction may be especially sensitive to how governance architectures organize coordinated adaptation over time. Institutional configurations characterized by stable coordination mechanisms, organizational continuity, and integrated sequencing authority may be better positioned to absorb disturbances internally, while fragmented or highly reopenable arrangements may face greater exposure to renegotiation and delay. In this sense, TCT is used here not as a universal explanation of nuclear construction outcomes but as an analytical framework for examining how such architectures may shape adaptive capacity under conditions of sustained uncertainty and interdependence.

2.3. Governance architectures and adaptive capacity

Building on this transaction-cost perspective, the central governance problem in nuclear construction is whether disturbances are absorbed through coordinated adaptation or externalized into renegotiation, procedural reopening, conflict escalation, and delay. Governance architectures refer here to the broader institutional configurations through which adaptation is organized across the nuclear construction process. They shape how authority is allocated, how coordination is organized across interdependent actors and how disturbances are managed throughout long implementation periods. These architectures are historically durable, but not fixed, and may evolve unevenly through regulatory reform, political realignment, institutional rupture, broader political and cultural shifts, and changing state–industry relations over time [25].

While nuclear construction duration is shaped by technology, shocks, and project-specific conditions, a TCT perspective suggests that governance architectures may condition whether disturbances are absorbed through coordinated adaptation or externalized into delay, renegotiation, and procedural reopening. Conceptually, this distinction is examined through two macro-institutional environments: Political Integration of Subnational Authority (PISA), referring to the degree to which subnational political authority is institutionally recognized and integrated into multi-level systems of political authority; and legal–regulatory proceduralism (LRP), referring to the extent to which oversight, approval, and dispute resolution are organized through formalized legal–regulatory procedures, independent institutions, and judicial or participatory mechanisms. These differences matter because they shape the capacity of nuclear governance architectures to coordinate adaptation under conditions of sustained uncertainty.

Adaptive capacity refers here to the ability of governance architectures to absorb regulatory, political, financial, and organizational disturbances through internally coordinated adaptation rather than prolonged renegotiation, procedural reopening, or delivery disruption. From this perspective, adaptive capacity is generated and sustained through interdependent mechanisms, conceptualized here as governance complexes. Rather than representing fixed institutional types, these complexes refer to recurring functional mechanisms through which governance architectures organize coordination and adaptation within nuclear delivery regimes. This follows a central proposition of TCT: under conditions of high asset specificity, uncertainty, and interdependence, adaptive capacity in nuclear construction is shaped by the alignment between institutional arrangements and the coordination requirements generated by transactional complexity [55,56,59].

3. Research design and methodology

This study employs an explanatory sequential mixed-methods design in which quantitative analysis is followed by qualitative comparative analysis [13,20]. The rationale for this approach is that variation in nuclear construction performance involves both measurable cross-national regularities and historically contingent governance dynamics that cannot be fully captured through statistical analysis alone. The quantitative strand identifies macro-institutional associations linked to variation in net construction lead time, while the qualitative strand provides explanatory elaboration of these patterns through a structured comparative analysis of secondary literature across different governance regimes and delivery systems. Integration between the two strands occurs through a sequential explanatory logic moving from macro-institutional association to comparative governance interpretation.

The quantitative strand examines all commercial nuclear reactors completed between 1954 and 2020 using reactor-level construction data derived from the IAEA Power Reactor Information System (PRIS) and compiled in the open-access Global NPP Database [7]. The final regression sample includes 541 completed reactors. The dependent variable is net construction lead time, measured as the elapsed time between construction start and commercial operation, excluding formally recorded suspension periods where applicable. Because construction durations are strongly right-skewed, the analysis uses the logarithm of net lead time in the regression models. Macro-institutional indicators are matched to the construction start year in order to approximate the broader macro-institutional environment within which project sequencing, licensing, financing, and implementation were originally organized. The quantitative strand operationalizes these macro-institutional dimensions using indicators derived from the V-Dem dataset [12]. PISA is measured using the Division of Power Index (v2x_feduni), while LRP is proxied by the Rule of Law Index (v2x_rule).

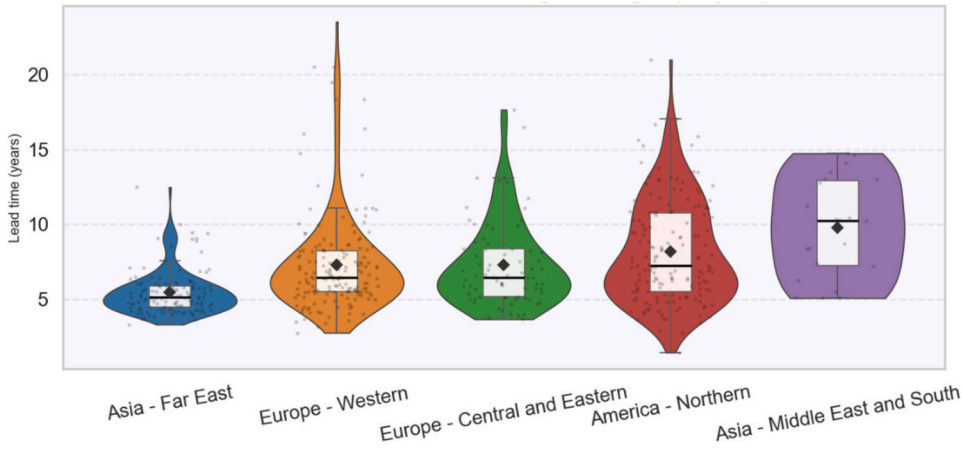


Fig. 1. Nuclear construction lead time by IAEA region, net years.

These indicators are treated as macro-institutional proxies for dimensions of governance architectures rather than as direct measurements. The regression analysis includes controls for global region fixed effects, construction decade fixed effects, log construction suspension days, energy-crisis exposure at project initiation, and post–Three Mile Island, Chernobyl, and Fukushima regulatory environments. Ordinary Least Squares (OLS) models are estimated using heteroskedasticity-robust standard errors and multiple fixed-effects specifications to account for temporal and regional heterogeneity. Within the explanatory sequential design, the quantitative strand serves a pattern-identification function by identifying recurring institutional associations and cross-national performance patterns that inform the subsequent qualitative analysis.

Building on these cross-sectional associations, the quantitative strand identifies recurring construction performance patterns through average construction duration and lead-time variability across major nuclear-building states that together account for most commercial reactor construction during the period under study. The analysis further distinguishes historically specific country-period regimes where governance arrangements evolved substantially over time. While the clusters are quantitatively derived, the comparative interpretation also considers important shifts in institutional arrangements and construction trajectories within countries where relevant for understanding changes in delivery performance. These outcome-based clusters provide the comparative structure for the subsequent qualitative analysis.

In the second, qualitative strand of the explanatory sequential design, evidence is examined through structured comparative analysis of secondary literature in nuclear governance, infrastructure studies, political economy, and energy transitions. This stage examines how similarities and differences in governance arrangements across clustered construction regimes relate to variation in delivery trajectories, adaptive capacity, and construction performance under conditions of sustained uncertainty and interdependence.

Finally, the two strands are integrated through an interpretive synthesis identifying four interdependent governance complexes as analytical abstractions of recurring functional mechanisms rather than predefined institutional categories. Together, these governance complexes provide a coherent interpretive lens for explaining how broader macro-institutional environments translate into variation in construction performance across nuclear delivery regimes.

4. Quantitative patterns and construction performance regimes

This section presents the quantitative strand of the analysis. It first examines the relationship between macro-institutional environments and variation in nuclear construction duration using reactor-level regression analysis. It then uses project-level trajectories and country-period clustering to identify construction performance regimes that guide the qualitative comparison strand developed in the next section.

4.1. Macro-institutional environments and construction duration

The analysis first examines broad patterns of variation in net construction lead times across the global reactor sample.

Fig. 1 shows substantial variation in nuclear construction duration across IAEA regions. Colored violins show the density of project-level construction durations; white boxes indicate medians and interquartile ranges; black diamonds mark regional means; and jittered dots represent individual reactor projects. The figure indicates that construction duration varies not only in mean levels but also in dispersion, motivating analysis of macro-institutional environments beyond technological or temporal factors alone.

To examine these macro-institutional associations systematically, the analysis estimates a series of Ordinary Least Squares (OLS) models in which the natural logarithm of net construction lead time is regressed on the two macro-governance indicators together with controls for construction suspensions, historical shock exposure, and regional and temporal fixed effects. Net construction lead time (LT_{net}) is first calculated in days and converted into months as:

$$LT_{net,i} = \frac{\text{Commercial Operation Date}_i - \text{Construction Start Date}_i - \text{Suspension Days}_i}{30.44}$$

The natural logarithm of this measure, $\ln(LT_{net,i})$, is used as the dependent variable in all regression models.

The full-control model specification is defined as:

$$\ln(LT_{net,i}) = \alpha + \beta_1 LRP_i + \beta_2 PISA_i + \beta_3 \ln(\text{days_susp}_i + 1) + \beta_4 \text{EnergyCrisis}_i + \beta_5 TMI_i + \beta_6 Chernobyl_i + \beta_7 Fukushima_i + \gamma_r + \delta_t + \varepsilon_i$$

where

Table 1
OLS regression results from full control model.

Variables†	Coefficient	(Robust SE)
PISA (v2x_feduni)*	−0.3640***	(0.104)
LRP (v2x_rule)	+0.4267**	(0.157)
ln(Construction suspension days)	+0.0280**	(0.010)
Energy crisis exposure (start-year)	−0.1137***	(0.034)
Three Mile Island exposure (post-event regime)	+0.4454***	(0.047)
Chernobyl exposure (post-event regime)	+0.3124***	(0.044)
Fukushima exposure (post-event regime)	+0.3321***	(0.053)
Constant	1.670***	(0.124)

Region fixed effects: Yes
Decade fixed effects: Yes
Observations: 541
R^2: 0.555
Estimator: OLS with HC1 robust standard errors.
† $p < 0.10$,
* $p < 0.05$,
** $p < 0.01$,
*** $p < 0.001$

- γ_r = IAEA regional fixed effects
- δ_t = decade fixed effects
- εi = stochastic error term

Table 1 reports the full-control OLS model for log net construction lead time. The results are consistent with the theoretical expectation that macro-institutional environments are associated with construction duration. Higher values of PISA (v2x_feduni) are associated with shorter construction durations, while higher levels LRP (v2x_rule) are associated with longer lead times. Substantively, moving from low to high values of PISA is associated with an approximately 30–35% reduction in expected construction duration, holding other factors constant, whereas higher levels of LRP are associated with an increase in expected duration of more than 50%.

Project-level controls behave as expected. Longer suspension periods are associated with significantly longer construction timelines, while energy-crisis exposure at project initiation is associated with shorter lead times, consistent with accelerated political prioritization and deployment urgency in post-crisis contexts. Projects initiated under post–Three Mile Island, Chernobyl, and Fukushima regulatory environments are associated with substantially longer construction durations, consistent with sector-wide regulatory tightening after major nuclear accidents.

Fig. 2. Estimated relationships between the two macro-institutional indicators and log net construction lead time. Panel A plots LRP (v2x_rule), and Panel B plots PISA (v2x_feduni). Points represent individual reactor projects (N = 541), while solid lines indicate model-estimated relationships from the full-control specification.

Fig. 3 reports the coefficient estimates from the full-control OLS model. Macro-institutional indicators are shown alongside construction

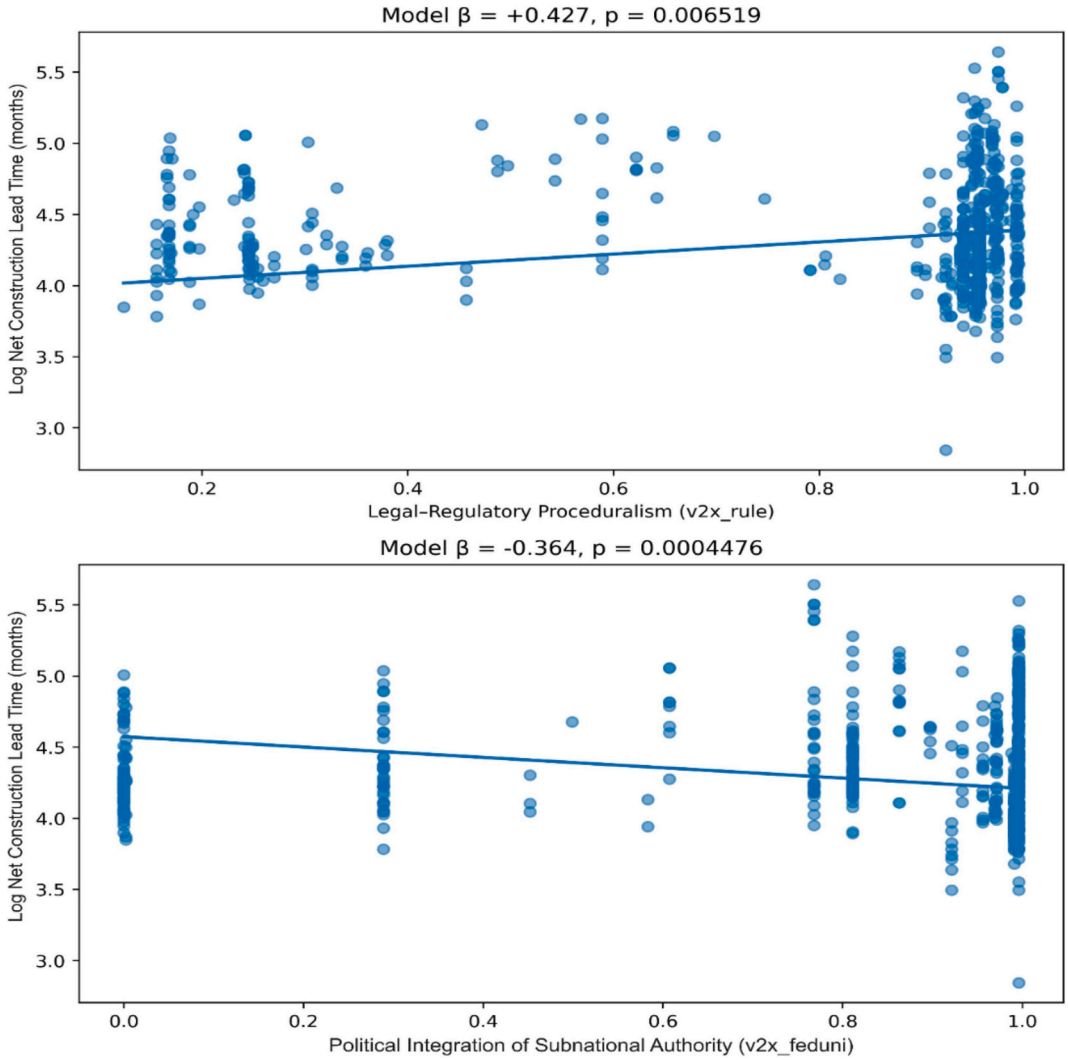


Fig. 2. Macro-Institutional Environments and Nuclear Construction Duration.

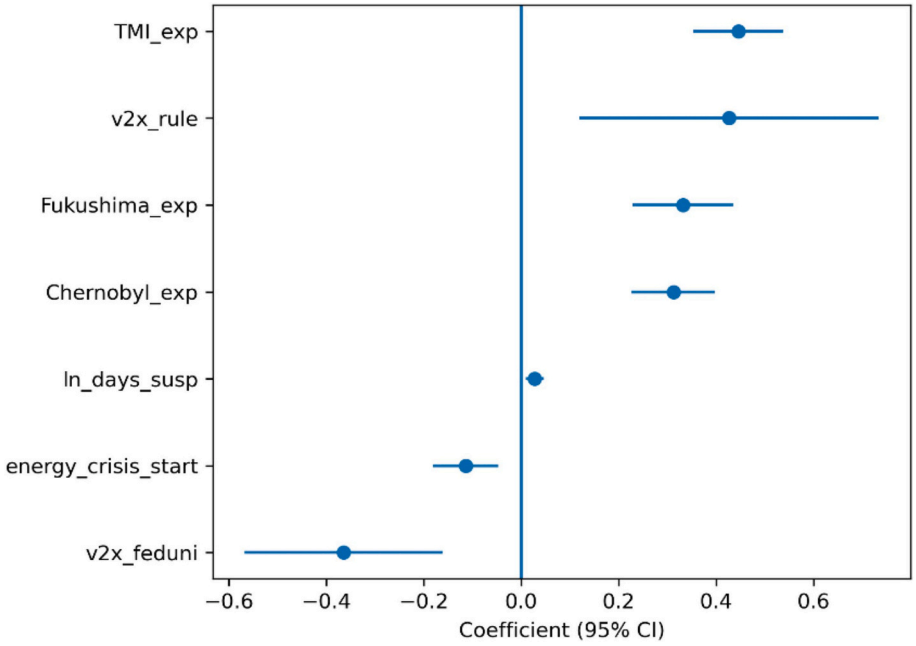


Fig. 3. Full-control OLS estimates for log net construction lead time, 95% CI.

suspension days, energy-crisis exposure, and major accident-period indicators. The figure shows that the two macro-institutional indicators remain sign-consistent and substantively robust when project disruptions, temporal shocks, and regional and decade fixed effects are included.

Fig. 4 shows that these institutional associations are not reducible to regional location alone. East Asian projects—especially China, South Korea, and pre-Fukushima Japan—cluster toward shorter construction durations, while North American projects exhibit longer lead times and higher LRP. Western European projects display greater dispersion, consistent with more mixed institutional configurations. Although regional histories matter, the visual pattern suggests that similar macro-institutional configurations may be associated with comparable construction durations across otherwise distinct national contexts.

Fig. 5 situates the institutional patterns within broader temporal dynamics. Construction durations rise across historical cohorts, reflecting increasing technical complexity and regulatory tightening after major safety accidents. Earlier cohorts display shorter and more compact lead times, while later cohorts show higher averages and greater dispersion. However, the broader macro-institutional patterns remain observable across decades.

Taken together, the regression results and visual patterns indicate that nuclear construction duration varies systematically across macro-institutional environments. The quantitative evidence does not imply that governance alone determines construction outcomes. Rather, it identifies recurring associations between macro-institutional conditions associated with different governance architectures and differences in construction lead-time outcomes. These patterns provide the empirical basis for the construction performance regimes identified in the next sections.

4.2. Construction performance regimes

To examine how the macro-institutional associations identified in the regression analysis relate to historically distinct delivery trajectories, the analysis focuses on major nuclear-building states that together account for most commercial reactor construction during the period under study. The selected country-period regimes capture substantial variation in construction performance, governance organization, and historical development paths across the global nuclear sector.[1] Rather than treating countries as internally uniform units, the analysis distinguishes historically specific construction regimes in which governance architectures changed substantially over time, allowing descriptive comparison of both delivery speed and delivery stability across different governance contexts.

Fig. 6 shows project-level construction trajectories across major nuclear-building countries. Three empirical patterns stand out. First, construction performance displays substantial country-period persistence rather than exhibiting simple convergence over time. Some regimes sustain relatively short and stable lead times over extended periods, while others display longer and more dispersed trajectories. Second, common external shocks are associated with heterogeneous national responses rather than uniform performance effects. Major accident periods and regulatory shifts coincide with sharper discontinuities in some contexts than in others. Third, countries differ not only in average construction duration but also in dispersion and tail risk, indicating that delivery stability is analytically distinct from delivery speed.

Fig. 7 positions country-period construction regimes according to mean net construction lead time and construction-duration variability. The horizontal axis captures mean lead time (years), while the vertical axis captures lead-time variability (standard deviation, years), used here as an inverse indicator of delivery stability. Together, these dimensions distinguish regimes that are fast and stable, moderately paced but controlled, slow and volatile, or structurally delayed but less dispersed.

Table 2 summarizes the outcome-based clusters visible in Fig. 7. Cluster 1, Fast and Stable, includes Japan before Fukushima and South Korea, both characterized by low average duration and low dispersion. Cluster 2, Intermediate Program Regimes, includes China, Sweden, France during its fleet-expansion period, Germany, and the USSR, combining moderate mean lead times with controlled variability.

[1] Canada is excluded because the qualitative comparison focuses on country-period cases selected to maximize variation across the main performance clusters and governance configurations.

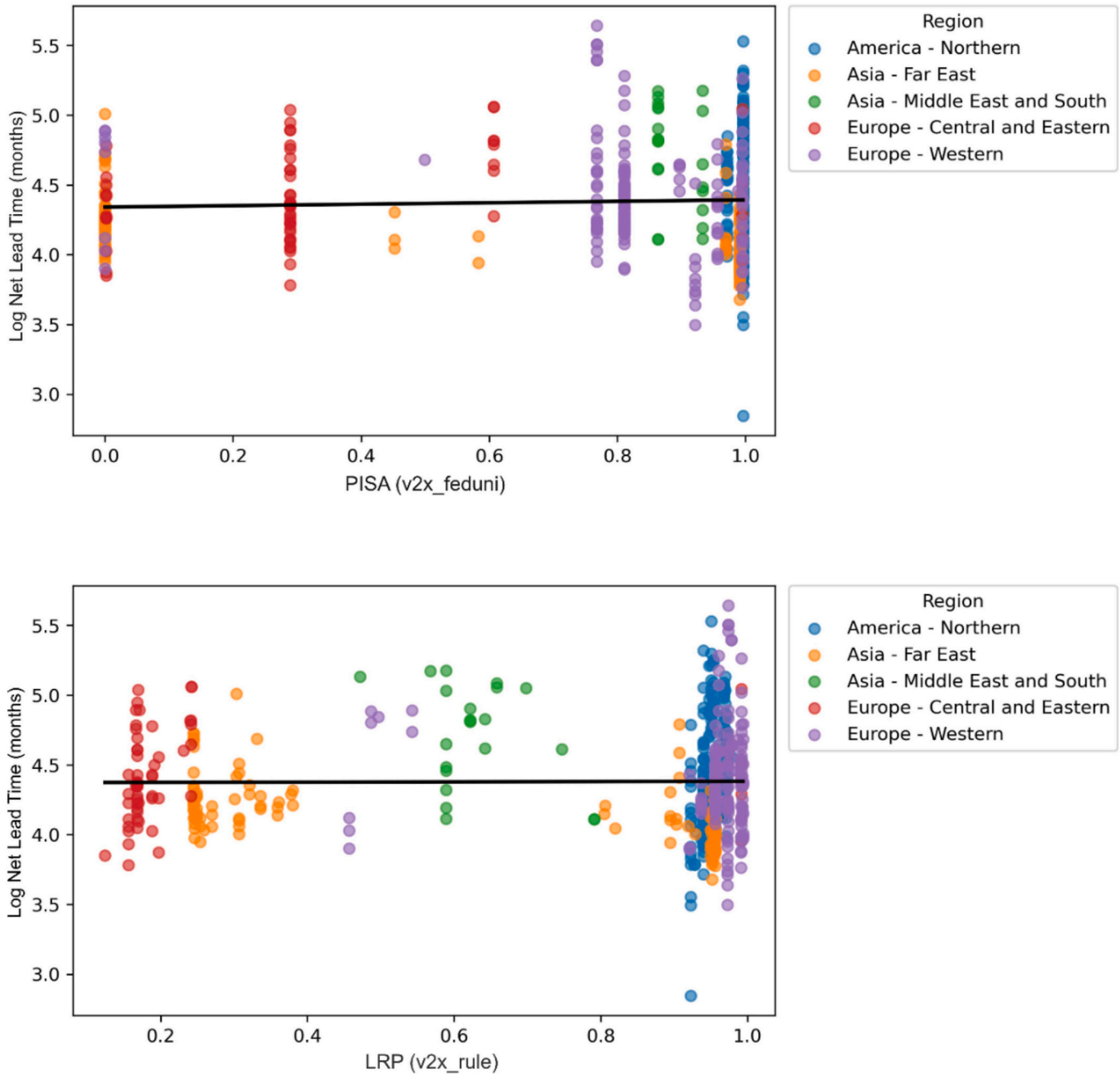


Fig. 4. Governance and nuclear construction duration by region.

Cluster 3, Slow and Volatile, includes the United States and the United Kingdom, where long mean lead times are combined with high dispersion and pronounced right tails. Cluster 4, Structural Outlier, includes the post-2006 Russian Federation, characterized by exceptionally long mean construction durations but comparatively moderate variability. Construction performance also varies over time within national contexts as institutional and organizational conditions evolve. France's later projects diverge from the earlier fleet-expansion trajectory, while the post-2006 Russian Federation differs substantially from the Soviet construction regime. These patterns suggest that construction outcomes are shaped not only by national context, but also by historically specific construction trajectories and institutional change over time.

These clusters are used as heuristic comparative categories rather than deterministic causal types. Their purpose is to organize observed variation in construction duration and delivery stability (as reflected in lead-time variability) and to identify the country-period regimes that require qualitative interpretation. The next section therefore examines how these performance regimes are associated with different governance architectures and adaptive mechanisms in practice.

5. Comparative governance regimes and nuclear delivery systems

Building on the construction performance regimes identified in Section 4, the qualitative analysis examines how different governance arrangements organize adaptation under conditions of uncertainty and interdependence. In particular, the comparative analysis examines how the macro-institutional associations identified quantitatively with PISA and LRP are organized and manifested across different governance architectures and delivery regimes. Rather than treating countries as

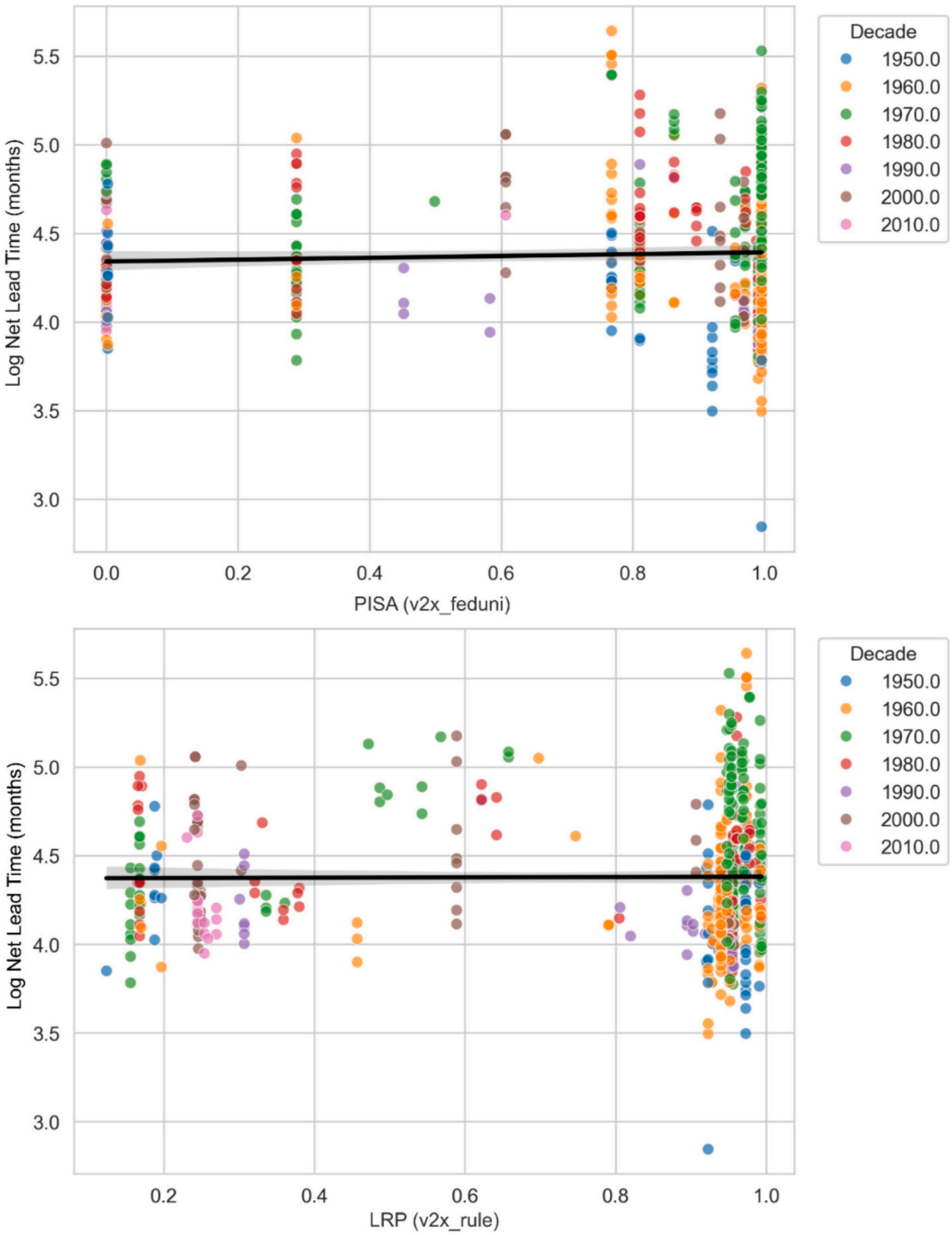


Fig. 5. Temporal patterns in governance structure and construction duration.

isolated case studies, the analysis develops a structured comparative interpretation of recurring governance patterns across the identified performance regimes, focusing on how multi-level sequencing authority, organizational continuity, financial stabilization, and political authorization are organized within different nuclear delivery systems. The regimes presented below therefore function as comparative governance ideal-types derived from historically recurrent configurations observed across the cases rather than as fixed national models.

5.1. Programmatic coordination regimes

5.1.1. (Japan pre-2011; South Korea)

Programmatic coordination regimes are characterized by politically and organizationally integrated forms of sequencing authority embedded within stable administrative–utility systems. Although formal decision-making authority may remain institutionally dispersed, coordination over siting, licensing, procurement, and construction sequencing is stabilized through durable organizational relationships and repeated deployment cycles. This supports standing program governance in which interface management, contractor coordination, and regulatory interaction are reproduced across successive projects rather than repeatedly renegotiated on a project-by-project basis. In Japan, coordination was stabilized through politically embedded administrative guidance and long-standing utility control over site development and construction scheduling, preserving construction continuity despite formally fragmented authority [28,35]. In South

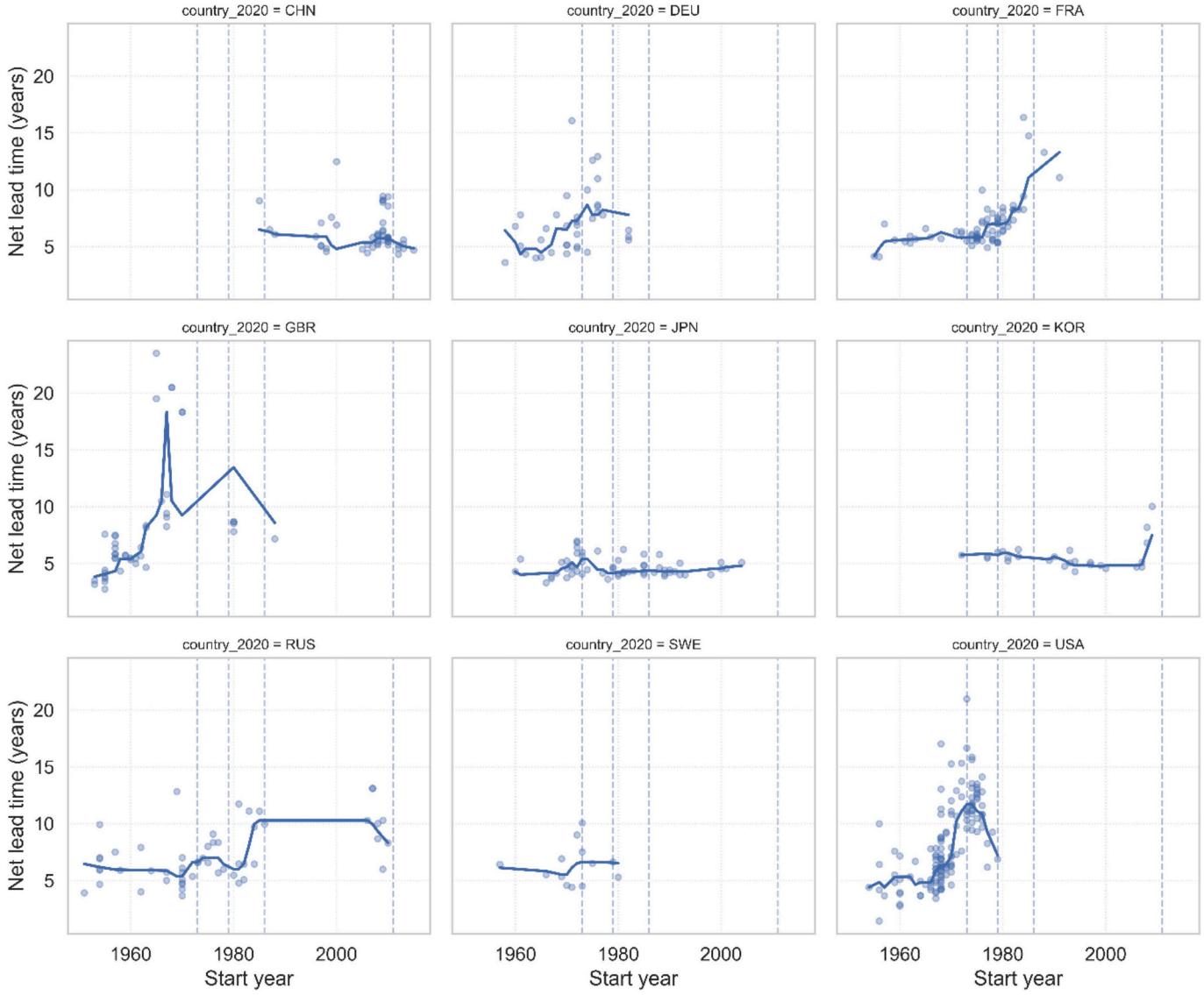


Fig. 6. Project-level nuclear construction lead times by construction start year and historical trajectories, 1955–2019.

Korea, vertically integrated utilities embedded within developmental-state coalitions enabled rapid internal resolution of interface problems and sustained batch deployment [33,51]. Unlike the more hierarchical expansion regimes discussed below, continuity here depended less on hierarchical administrative steering than on politically durable organizational networks and sectoral coalitions.

These arrangements also supported cumulative technical learning and long-horizon financial continuity. Standardized deployment patterns, stable supplier qualification, and routinized regulatory interfaces allowed knowledge and coordination routines to accumulate across projects rather than being repeatedly reconstructed [28,48]. Investment risk was largely internalized within utility-centered or state-linked systems, while legitimacy during expansion phases rested on technocratic and developmental narratives combined with politically mediated forms of subnational authority incorporation [23,28]. As a result, disturbances were more often absorbed through incremental adjustment than translated into institutional reopening, financing disruption, or delivery instability. The post-2011 Japanese experience nevertheless illustrates that this stability remained contingent on the durability of political authorization arrangements, as rising judicial and local contestation increased institutional reopenability even where technical and industrial capabilities remained strong [29].

Together, these regimes combined embedded coordination, cumulative learning, and internalized adaptation associated with shorter and less variable construction durations.

5.2. *Hierarchically coordinated expansion regimes*

5.2.1. *(France fleet era; China; Sweden build phase; Germany; USSR)*

Hierarchically coordinated expansion regimes are characterized by vertically integrated sequencing authority embedded within hierarchical planning bodies, state-owned enterprises (SOEs), or corporatist utility systems. Rather than relying on negotiated project-by-project coordination, these regimes organized reactor construction through national rollout strategies, administrative steering, and sectorally integrated planning arrangements that stabilized sequencing authority during active expansion phases. In France's fleet era, technocratic coordination integrated EDF, CEA, and Framatome within a dirigiste state framework that concentrated site selection, procurement, and construction sequencing [36,54]. In China, vertically integrated SOEs operating under NDRC and SASAC oversight coordinated reactor standardization, financing approval, and multi-unit site sequencing through national deployment programs [1,48]. France's post-fleet experience at Flamanville-3 [43] illustrates how these arrangements may evolve over

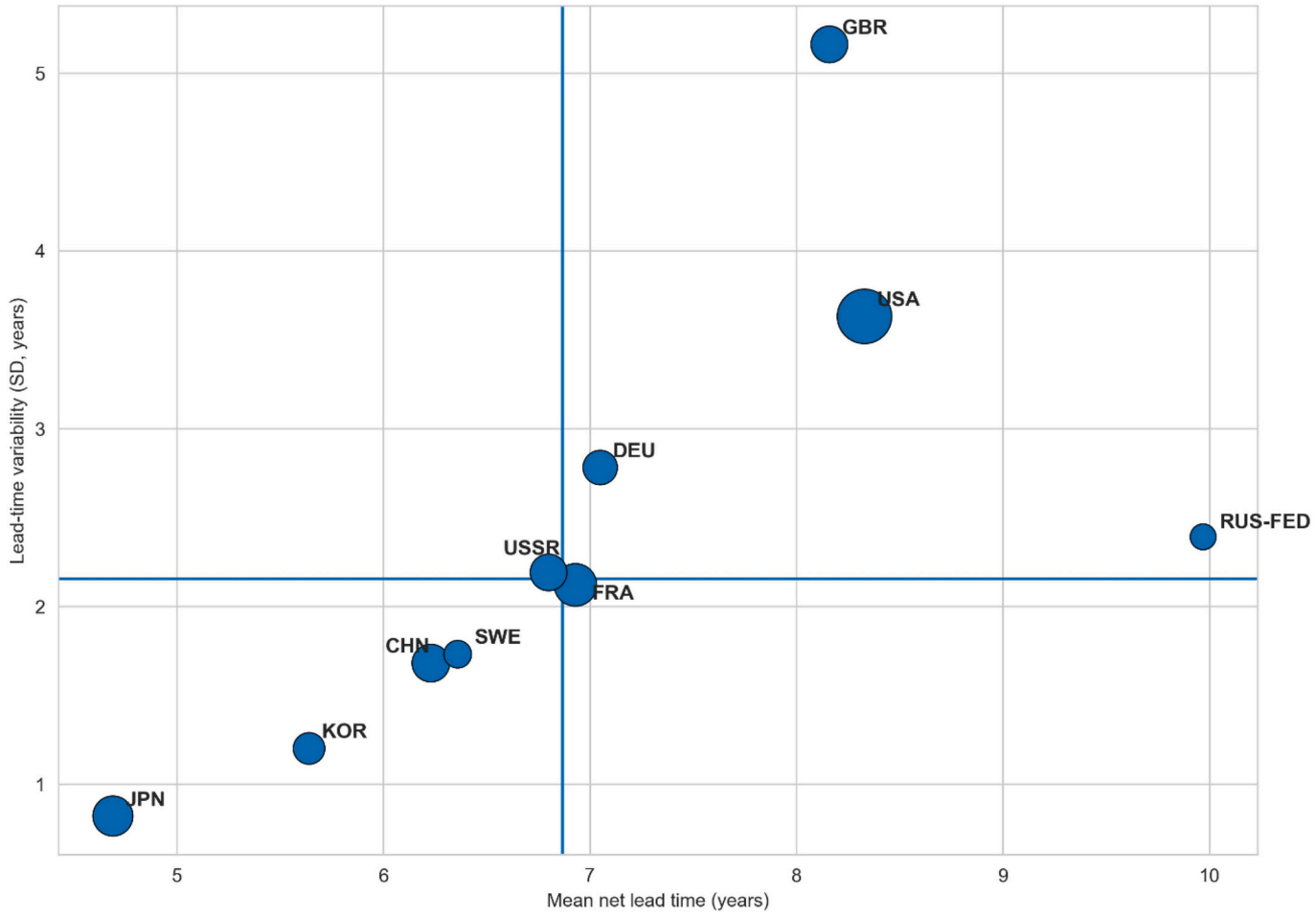


Fig. 7. Cross-national dispersion in construction duration and delivery stability.

Table 2
Outcome-based performance clusters of nuclear construction.

Cluster	Mean Lead Time (years)	Lead-Time Variability (SD, years)	Countries	Interpretation
1. Fast & Stable	Mean < 6	SD < 2	Japan, South Korea	Low average duration and low dispersion
2. Intermediate Program Regimes	6 ≤ Mean < 7	SD < 3	China, Sweden, France, USSR, Germany	Moderate duration with controlled variability
3. Slow & Volatile	8 ≤ Mean < 9	SD > 3	United Kingdom, United States	Long and highly variable delivery
4. Structural Outlier	Mean ≥ 9	SD < 3	Russia-Fed (post-2006)	Exceptionally long construction durations

time toward more procedurally reopenable and legally contestable governance conditions consistent with higher LRP, helping explain the sharp divergence from earlier fleet-era construction performance.

In Sweden, corporatist planning institutions and public ownership concentrated coordination authority within utilities such as Vattenfall during the build phase, combining hierarchically coordinated siting and workforce planning with negotiated incorporation of subnational actors rather than adversarial procedural contestation [21]. In the USSR, ministerial command structures embedded design bureaus and construction trusts directly within state planning systems, treating reactor deployment as industrial production rather than as isolated project delivery [26,27]. Germany similarly stabilized sequencing authority through corporatist utility–state coordination during the expansion phase, with Länder governments, utilities, and federal actors linked through negotiated planning arrangements despite formal federal dispersion [22,54]. Unlike the programmatic coordination regimes discussed above, continuity here depended less on embedded utility networks than on hierarchical or corporatist steering structures capable of sustaining large-scale expansion programs.

These arrangements also supported cumulative industrial learning, standardized replication, and relatively stable long-horizon financing conditions. Integrated design authority and repeated deployment cycles enabled transfer of routines, preservation of supplier coordination, and accumulation of organizational capability across projects, although this did not necessarily translate into declining costs or linear efficiency gains, as evidenced by negative learning in the French fleet experience [1,18,36,60]. In Sweden and the USSR, corporatist utilities and state planning bodies stabilized contractor relations and workforce deployment, embedding learning within industrial organizations rather than isolated project teams [21,27]. Financial continuity was likewise internalized through public ownership, state planning, or administratively coordinated investment frameworks rather than market-based project finance. In France and China, construction and revenue risks were largely absorbed within state-linked systems governed through rollout logic rather than project-level bankability [1,36]. In Germany and Sweden, utility-centered investment structures initially buffered construction risk, although later politicization increased perceptions of

reversibility despite retained technical capability [21,41,54]. Legitimacy during expansion phases rested primarily on technocratic authority, corporatist settlement, or state-centered developmental narratives, limiting institutional reopenability during construction while leaving long-term continuity vulnerable to broader political reorientation [15,19,37,48]. France and China largely absorbed regulatory adjustments without reopening program mandates, whereas Sweden and Germany eventually translated rising contestation into political exit trajectories rather than continuous procedural destabilization of ongoing construction [21,41]. In the USSR, continuity persisted until systemic political rupture disrupted both legitimacy and organizational authority [26,27].

These regimes show that hierarchical coordination, industrial replication, and state-backed continuity were associated with stable construction performance during expansion, while remaining vulnerable to political realignment.

5.3. Fragmented project-based regimes

5.3.1. (United States; United Kingdom)

In the United States, the fragmented governance configuration that later characterized nuclear construction emerged gradually after the breakdown of the earlier Atomic Energy Commission model, which combined promotional and regulatory authority within a relatively administratively integrated delivery regime. The creation of the Nuclear Regulatory Commission, together with the expansion of NEPA-based hearings and judicial review, institutionalized a more legally and administratively contestable system marked by fragmented oversight, expanded opportunities for regulatory reopening, and increasingly layered scrutiny during project development [40,52]. In the United Kingdom, nuclear governance similarly evolved from an earlier state-led and technocratically coordinated regime organized through the UK Atomic Energy Authority and nationalized electricity institutions toward a more publicly and procedurally contestable system shaped by increasingly extensive public inquiries, expanding regulatory scrutiny, and layered authorization processes [23,42].

Fragmented project-based regimes are characterized by dispersed sequencing authority **across** regulators, courts, utilities, financial actors, and multiple political principals, such that no single organization maintains durable control over long-term program coordination. Despite formal recognition, subnational political authority is only weakly incorporated into durable delivery systems, shifting conflict processing toward procedural and judicial arenas. In the United States, authority over siting, licensing, financing, and oversight became fragmented across federal regulators, state utility commissions, courts, and private utilities, requiring coordination structures and financial arrangements to be reassembled for each project cycle [4,24]. In the United Kingdom, privatization and arm's-length regulation similarly organized delivery through project-specific coalitions supported by planning reforms and ad hoc policy instruments rather than standing national programs [4,5]. Delivery therefore proceeded through bespoke contractual and regulatory arrangements rather than cumulative fleet-based deployment, raising coordination and enforcement costs under uncertainty.

These institutional conditions also undermine cumulative learning and long-horizon continuity. Repeated interruptions weaken design stability, destabilize supplier networks, and erode workforce continuity, recreating organizational FOAK (first-of-a-kind) conditions even where reactor technologies are mature [3,4]. In both the United States and the United Kingdom, isolated projects depend heavily on temporary EPC consortia and discontinuous project teams, limiting the accumulation of routinized interface management and construction experience. Financial arrangements further reinforce these conditions. Construction and revenue risks are largely externalized to private sponsors and investors, while revenues depend on market or quasi-market arrangements vulnerable to price volatility, policy revision, and political reopening [3,4]. As a result, technical setbacks and cost escalation are more likely to trigger renegotiation, redesign of support mechanisms or project suspension, rather than being absorbed through internally coordinated adaptation. Legitimacy is similarly organized through procedural–adversarial channels rather than negotiated political incorporation, embedding political contestation within courts, regulatory hearings, and dispersed veto structures [4,6,52]. Disturbances therefore more frequently translate into institutional reopening and policy redirection rather than coordinated adaptation.

Overall, these regimes combine fragmented coordination, discontinuous learning, market-exposed finance, and procedurally reopenable authorization associated with long and highly variable construction durations despite substantial industrial capability.

5.4. Reconstructed state-corporate recovery regimes

5.4.1. (Post-2006 Russian Federation)

Reconstructed state-corporate recovery regimes are characterized by the reconstruction of delivery coordination following systemic institutional rupture through renewed organizational concentration of sequencing authority within state–corporate structures. In the post-Soviet Russian Federation, authority over ownership, regulation, investment, and project sequencing was progressively consolidated within Rosatom as a state corporation combining industrial and strategic functions [38]. Following the fragmentation and disruption of the 1990s transition period, sequencing authority and interface management were increasingly reorganized through internal corporate hierarchies rather than dispersed contractual coordination, thereby reducing inter-organizational bargaining and restoring partial program continuity [44]. Delivery subsequently proceeded through portfolio-based domestic and export programs that stabilized construction performance relative to the immediate post-Soviet period, although without fully recovering the delivery speed associated with the earlier Soviet expansion regime [31].

These arrangements also supported partial reconstruction of industrial learning, financial continuity, and political authorization. Consolidated sectoral control over certification, reactor design portfolios, and project deployment enabled renewed standardization and replication across domestic and export reactor series [38]. Supply chains and engineering capabilities were progressively reintegrated within state-corporate conglomerate structures, allowing interface routines and organizational knowledge to accumulate across projects rather than being repeatedly rebuilt from scratch [44]. At the same time, workforce losses and generational capability gaps inherited from the post-Soviet transition constrained productivity gains and contributed to persistently elevated construction durations [31]. Financially, sovereign-backed portfolio governance bundled financing, construction, fuel services, and long-term contractual arrangements within state-supported packages that reduced exposure to private capital markets and limited exit risk [44]. Construction and revenue uncertainty were therefore largely internalized within state-corporate structures, even where opacity and discretionary political buffering partially substituted for market discipline [31,38]. Political authorization similarly relied on reconstructed strategic-sector governance and containment of procedural contestation within executive and administrative structures rather than judicialized reopening of approvals [27,44]. This reduced variance relative to more fragmented regimes, while preserving relatively long lead times associated with post-transition state-corporate reconstruction.

This regime illustrates how coordination capacity, industrial continuity, and delivery stability can be partially rebuilt after systemic disruption, even where reconstruction costs constrain construction speed.

Table 3
Governance Complexes and Expected Performance Implications in Nuclear Construction.

Governance Complex	Core Function	Institutional Features	Comparative Evidence	Expected Performance Implications
Coordination & Authority (C1)	Stabilizes sequencing authority and interface management across long construction **periods.**	Programmatic coordination; vertically integrated utilities or SOEs; stable siting and licensing authority; routinized regulator–industry interaction; continuity across projects	South Korea and pre-2011 Japan show highly stable sequencing and fleet continuity; France's fleet era and Sweden's build phase demonstrate hierarchically or corporatist coordinated delivery; the USSR and China illustrate hierarchical coordination through state planning structures; the US and UK illustrate fragmented project-based coordination	Greater coordination continuity is associated with shorter and more stable construction durations; fragmented authority increases renegotiation, interface conflict, and delivery volatility
Technical Learning (C2)	Converts repetition into cumulative organizational and industrial capability	Standardized reactor series; stable EPC and supplier networks; retained workforce pipelines; routinized regulatory interfaces; continuity in design and construction practices	France's standardized fleet deployment; South Korean serial replication; Chinese multi-unit build programs; Sweden's coordinated utility-led build phase; USSR design-bureau continuity; contrast with discontinuous learning in the US and UK	Sustained repetition reduces redesign and requalification costs, improving delivery stability; interrupted programs recreate organizational FOAK conditions and increase delay risk
Risk & Finance (C3)	Determines whether uncertainty is internally absorbed or externalized into financeability crises	State-backed finance; utility-based risk internalization; long-term revenue stabilization; sovereign guarantees; durable policy commitment	France, China, Sweden, and the USSR internalized risk within public or corporatist systems; post-2006 Russia reconstructed sovereign-backed portfolio governance; the US and UK relied more heavily on market-based finance and project-specific support mechanisms	Internalized risk structures support continuity despite cost escalation; externalized financial exposure amplifies renegotiation, investor withdrawal, and project instability
Legitimacy & Institutional Trajectory (C4)	Shapes whether political and regulatory shocks are absorbed, reopened, or translated into institutional rupture	Technocratic legitimacy; corporatist incorporation; procedural contestation; institutional reopenability; shock-response architecture	Japan prior to Fukushima and South Korea relied on stable developmental legitimacy; France and Sweden incorporated nuclear expansion within corporatist political settlements; Germany and Sweden later shifted toward political exit trajectories; the US and UK institutionalized procedural contestation; post-Soviet Russia pursued re-centralized state-corporate authorization	Stable authorization arrangements reduce reset risk during construction; procedurally reopenable systems increase exposure to litigation, redesign, and political reversal; abrupt legitimacy shifts can disrupt or terminate otherwise stable delivery trajectories.

6. Interdependent governance complexes

The comparative analysis suggests that variation in nuclear construction performance reflects recurring combinations of governance functions that shape how delivery systems organize adaptation under uncertainty, interdependence, and long project horizons. Across the governance regimes identified in Section 5, differences in construction duration and delivery stability are consistently associated with how sequencing authority, technical learning, risk allocation, and political authorization are organized and coupled over time.

Rather than representing fixed institutional models, the governance complexes developed below are analytical abstractions derived from recurring patterns across cases. Together, they describe interdependent governance functions that shape whether disturbances are absorbed through coordinated adaptation or externalized into renegotiation, procedural escalation, and delay. The comparative evidence further suggests patterned relationships between these governance complexes and the macro-institutional dimensions associated with PISA and LRP, indicating that broader macro-institutional environments condition adaptive capacity and delivery performance over time. The four governance complexes identified across the comparative analysis are summarized in Table 3.

6.1. Coordination & authority complex (C1)

The Coordination & Authority Complex captures how sequencing authority and interface management are organized across long nuclear construction cycles. The central issue is not whether governance systems are simply more or less centralized, but whether planning, licensing, procurement, financing, and construction sequencing are coordinated within stable delivery structures or repeatedly recomposed across autonomous actors. Under conditions of high asset specificity, organizational interdependence, and long implementation horizons, fragmented authority increases coordination and renegotiation costs, making adaptive adjustment more difficult when disturbances occur.

Comparative evidence indicates that shorter and more stable construction durations are consistently associated with programmatic forms of coordination in which sequencing authority remains embedded within durable administrative, utility, or state-corporate arrangements. South Korea and pre-2011 Japan sustained continuity through stable utility-centered coordination and routinized regulator–industry interaction [28,30,33,35,51]. France's fleet-expansion period, Sweden's build phase, China's state-led rollout, and the Soviet program similarly stabilized sequencing authority through corporatist utilities, hierarchical planning systems, or vertically integrated SOEs [1,21,26,27,48,54]. By contrast, the United States and United Kingdom relied on project-specific coalitions assembled through dispersed regulatory, financial, and political bargaining, increasing exposure to procedural escalation and delivery disruption [4,5,24].

Functionally, C1 reflects the interaction of three reinforcing dimensions: stabilized authority over project sequencing, continuity in interface management across organizations, and sustained construction rhythms across successive builds. Where these elements remain aligned, adaptation is more likely to occur through internal coordination rather than repeated contractual and regulatory recomposition. This interpretation is consistent with the econometric finding that higher PISA levels are associated with shorter construction durations. Importantly, the comparative evidence also indicates that similar coordination outcomes can emerge through different institutional forms—including corporatist utility systems, developmental-state utilities, hierarchical planning

structures, and vertically integrated SOEs. What appears most consequential is therefore not institutional form itself, but the continuity and functional integration of sequencing authority over time.

6.2. Technical learning complex (C2)

The Technical Learning Complex captures whether delivery systems can accumulate and retain technical and organizational capability across successive construction cycles. Whereas the Coordination & Authority Complex (C1) concerns the organization of sequencing authority, C2 concerns whether repetition is sustained and converted into routinized execution rather than recurring FOAK conditions. In nuclear megaprojects, technical learning is therefore not only an engineering process but also an institutional one, depending on governance arrangements capable of stabilizing design baselines, supplier interfaces, workforce continuity, and regulatory interaction over time.

Comparative evidence indicates that shorter and more stable construction trajectories are consistently associated with sustained industrial repetition and organizational continuity. France's fleet-expansion period, South Korea's serial deployment model, China's multi-unit build programs, Sweden's coordinated utility-led expansion phase, and Soviet-era standardization all supported cumulative learning through standardized reactor series, supplier networks, and repeated regulator–industry interaction [1,21,27,33]. Under these conditions, interface routines, construction practices, and tacit organizational knowledge were stabilized across projects. By contrast, the United States and United Kingdom experienced more discontinuous construction trajectories in which interrupted deployment cycles, fragmented EPC arrangements, and unstable supplier pipelines repeatedly recreated organizational first-of-a-kind conditions despite technological maturity [3,4,40,42,52].

Functionally, C2 reflects the interaction of three reinforcing dimensions: continuity in design and standardization regimes, stability in industrial and regulatory interfaces, and retention of workforce and organizational memory across projects. Where these elements remain aligned, learning becomes cumulative and adaptation costs decline as routines stabilize across projects. Where continuity is repeatedly disrupted, delivery systems lose the capacity to preserve accumulated capability, increasing redesign, requalification, and coordination burdens. This interpretation is consistent with the broader comparative finding that nuclear construction performance does not converge through technological maturity alone. Similar reactor technologies can exhibit different delivery trajectories depending on whether governance systems sustain repetition and organizational continuity over time.

6.3. Risk & finance complex (C3)

The Risk & Finance Complex captures how construction, revenue, and policy uncertainty are allocated across the long investment horizons characteristic of nuclear megaprojects. In this context, finance operates not only as a funding mechanism but also as a governance mechanism shaping whether disturbances are absorbed through coordinated adjustment or escalate into renegotiation, redesign, and interruption. The central issue is therefore not simply the availability of capital, but whether governance arrangements sustain credible long-term commitment under conditions of cost escalation, regulatory change, and political uncertainty.

Comparative evidence indicates that faster and more stable construction trajectories are consistently associated with institutional arrangements that internalize construction and revenue risk within durable public, utility, or state-linked systems. France's fleet-expansion model, China's state-directed rollout, Sweden's utility-centered build phase, and Soviet-era planning structures all relied on relatively stable financing environments in which investment continuity was supported through public ownership, administratively coordinated rollout strategies, or long-term policy commitment [1,21,27]. Post-2006 Russia similarly reconstructed sovereign-backed portfolio financing through Rosatom's vertically integrated state-corporate model [38,44]. Under these arrangements, cost overruns and schedule extensions were more likely to be managed through internally coordinated adjustment and continued political commitment rather than immediate project abandonment. By contrast, the United States and United Kingdom relied on project-based financing arrangements exposed to market volatility, regulatory uncertainty, and reopenable political support mechanisms, increasing the likelihood that technical disruptions would translate into financing crises and renegotiation cycles [4,6,24].

Functionally, C3 reflects the interaction of three reinforcing dimensions: the allocation of construction and schedule risk, long-term revenue and cost-recovery stability, and the perceived durability of political and regulatory commitment. Where these dimensions remain aligned, governance systems are better positioned to absorb uncertainty without destabilizing project continuity. Where they diverge, disturbances are more likely to translate into contractual conflict, investor withdrawal, and delivery disruption. This interpretation is consistent with the broader comparative pattern that slow and volatile construction trajectories emerge not only from technical complexity, but from governance arrangements that externalize uncertainty into market, regulatory, and political renegotiation processes.

6.4. Legitimacy & institutional trajectory complex (C4)

The Legitimacy & Institutional Trajectory Complex captures how political authorization is produced, stabilized, and reconfigured across long nuclear construction cycles. In this context, legitimacy shapes whether foundational decisions regarding siting, licensing, financing, and program continuation remain durable during construction or become increasingly institutionally reopenable under changing political and regulatory conditions. The central issue is therefore not whether conflict exists, but how governance systems organize and process political contestation over time.

Comparative evidence indicates that shorter and more stable construction trajectories are associated with governance arrangements in which political authorization is embedded within relatively durable technocratic, corporatist, or developmental coalitions. Pre-2011 Japan and South Korea sustained legitimacy through developmental and technocratic narratives combined with stable utility-centered coordination structures [23,30,33]. France's fleet-expansion period and Sweden's build phase similarly incorporated nuclear expansion within corporatist political settlements that limited procedural reopenability during construction [21,24]. China relied on hierarchical administrative authority and performance-based legitimacy within a state-led development framework [1,48]. By contrast, the United States and United Kingdom institutionalized procedurally adversarial forms of legitimacy in which contestation was processed through courts, regulatory hearings, and fragmented veto structures, increasing the likelihood that disturbances translated into redesign, delay, or policy reversal [4,6,24,53]. Post-Soviet Russia represents a distinct trajectory in which legitimacy and delivery continuity were reconstructed through political and organizational re-centralization of strategic-sector governance following systemic rupture [38,44].

Functionally, C4 reflects the interaction of three reinforcing dimensions: how legitimacy is produced and maintained, the degree of institutional reopenability surrounding prior commitments, and how governance systems respond to political and regulatory shocks. Where legitimacy is incorporated within relatively stable administrative or negotiated frameworks, including durable political incorporation of subnational authority, adaptation is more likely to occur through incremental adjustment without dismantling delivery continuity. Where contestation remains highly reopenable through procedural and judicial mechanisms, the risk of institutional reset rises and disturbances are more likely to escalate into broader disruption. Importantly, legitimacy does not operate independently from the other governance complexes.

The comparative evidence suggests that shocks are more likely to be absorbed where coordination capacity (C1), learning continuity (C2), and financial commitment (C3) remain intact to sustain coordinated adaptation over time.

6.5. Synthesis

Taken together, the four governance complexes suggest that nuclear construction performance depends less on isolated policy instruments or regime types than on how governance systems sustain coordinated adaptation under conditions of uncertainty and long project horizons. Across cases, variation in delivery speed and stability reflects the alignment—or misalignment—of interdependent governance functions related to coordination, learning continuity, risk absorption, and political authorization over time. The comparative evidence further suggests that these governance complexes exhibit patterned relationships with the broader macro-institutional dimensions associated with PISA and LRP, indicating that broader macro-institutional environments condition how governance systems organize adaptive capacity and nuclear construction performance over time.

Fig. 8 synthesizes the paper's interpretive framework by illustrating how broader macro-institutional environments shape construction performance through the four governance complexes identified in the comparative analysis. The joint configuration of PISA and LRP, reflecting their institutional co-evolution, conditions these governance complexes, whose alignment reflects the system's capacity to absorb uncertainty and helps explain variation in average construction duration and delivery stability.

7. Discussion and conclusion

Recent research on infrastructure megaprojects and nuclear construction has shown that delivery performance is highly sensitive to governance dynamics, particularly regulatory evolution, institutional instability, and the capacity to coordinate adjustment under uncertainty [10,47,55,59]. What remains less clearly specified is why some governance systems sustain relatively stable construction trajectories while others experience chronic delay, volatility, and discontinuity despite operating under comparable technological and financial conditions.

This paper addressed that question through a mixed-methods analysis combining global regression evidence with comparative historical interpretation. The quantitative results indicate that shorter construction durations are systematically associated with macro-institutional environments characterized by greater political integration of subnational authority, while longer and more volatile lead times are associated with more procedurally reopenable and judicialized institutional settings. These relationships should not be interpreted as direct measures of governance architecture itself, but rather as broad institutional correlates that condition how adaptation is organized during reactor construction.

The comparative qualitative analysis further suggests that construction performance depends less on individual project characteristics

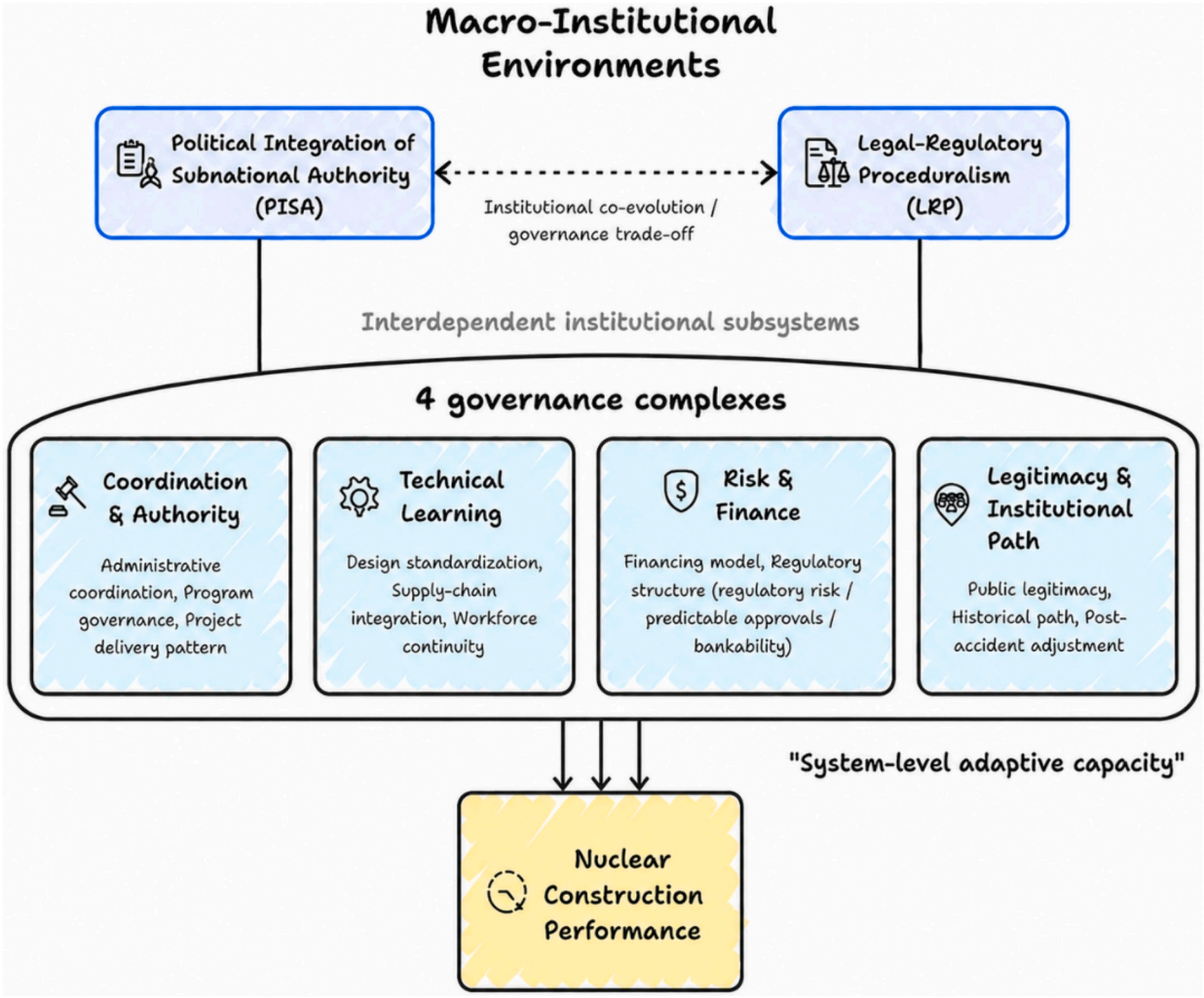


Fig. 8. Macro-institutional environments, governance complexes, and nuclear construction performance.

than on whether governance systems sustain continuity across successive build cycles. Across the identified performance regimes, delivery outcomes vary according to how coordination authority, technical learning, financial commitment, and political authorization are organized and coupled over time. In some settings, disturbances such as regulatory change, financial stress, or political contestation are absorbed through program-level adjustment and organizational continuity. In others, similar disturbances are more likely to translate into renegotiation, institutional reopening, and delivery disruption.

From this perspective, the paper's primary contribution is analytical and interpretive rather than predictive. Rather than proposing a deterministic model of nuclear construction performance, the analysis develops four governance complexes as analytical abstractions that help explain recurring cross-national patterns identified through quantitative and qualitative integration. The findings therefore complement governance-centered approaches to megaproject performance by suggesting that adaptive capacity—the institutional capability to coordinate adjustment, preserve continuity, and manage conflict across long implementation horizons—represents a critical dimension of infrastructure delivery performance.

The analysis also helps explain why construction outcomes do not converge across countries despite shared reactor technologies, global supply chains, and accumulated engineering experience. Similar technological systems can operate either as cumulative long-term programs or as fragmented sequences of projects, depending on whether governance arrangements sustain stable sequencing authority, organizational learning, credible commitment, and institutional continuity over time. In this sense, nuclear construction functions not only as a technological challenge but also as a demanding test of long-horizon governance capacity under persistent uncertainty.

Several limitations should nevertheless be acknowledged. First, the quantitative analysis relies on institutional proxies derived from the V-Dem dataset and includes only completed reactors, excluding cancellations and abandoned projects. This means the analysis captures broad institutional environments rather than governance architectures directly and may understate governance dysfunction in regimes characterized by high abandonment rates, particularly fragmented project-based systems. The identified relationships should therefore be interpreted cautiously rather than as precise causal measurements of specific governance mechanisms. Second, the cluster construction and comparative synthesis necessarily simplify historically heterogeneous national trajectories into heuristic performance regimes and governance complexes, potentially obscuring important within-country variation and limiting causal precision. Finally, while the historical scope of the study strengthens comparative breadth, contemporary nuclear programs may evolve under new geopolitical, industrial, and regulatory conditions that differ from earlier construction eras.

Future research should further investigate how governance systems sustain coordinated adaptation over long implementation horizons under evolving geopolitical, industrial, and regulatory conditions. Extending the mixed-methods approach developed here to contemporary nuclear programs and other large-scale infrastructure sectors may help clarify how institutional continuity, political contestation, financial pressures, supply-chain constraints, decarbonization pressures, and evolving state–market relationships interact to shape delivery performance over time.

CRediT authorship contribution statement

Jorge Ernesto Rodríguez-Morales: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mikael Höök:** Writing – review & editing, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mikael Hook reports financial support was provided by Swedish Energy Agency. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.erss.2026.104858.

Data availability

Data will be made available on request.

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